

Receiver-Relative Arithmetic Configuration Calculus

An exact ecological carrier for perspective, boundary, prime incidence, formulation transport, and higher-order change

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Abstract. This paper defines the missing calculus suggested by the laboratory’s receiver, prime, transcendental-formulation, and dynamic-system research. Its base is the actual incidence of co-present receiver lineages rather than their full Cartesian product. A joint receiver and typed comparison laws generate the contemporary accessible region, its boundary strata, and its discriminant. A receiver partial exists only when that incidence admits the corresponding tangent; entangled perspectives are not supplied false independent coordinates. Arithmetic accessibility is retained as a chain-valued valuation series, so prime axes, prime squares, mixed semiprimes, multiplicity, orientation, and wall crossings remain present before a count is taken. The ordinary covariant derivative and exact discrete jump are then two parts of one stratified fundamental theorem. At second order, joint incidence is separated from curvature: a symmetric mixed difference can be nonzero even when the two transport orders commute. The finite prime interval from 113 to 127 is reconstructed as six exact valuation-wall events hidden by the endpoint gap 14. The same carrier is applied to factorial, arctangent, Gaussian, Machin, and Chudnovsky formulations, showing that a transcendental invariant has an evaluation fiber of nonidentical arithmetic ecologies rather than one coefficient taxonomy. The result is an agnostic but lawful configuration calculus: each ecology supplies its incidence, boundaries, transports, and decorations; the common mathematics supplies restriction, transport, seam, return, mixed difference, and exact rebase. Its Riemann-Hypothesis relevance is precise. Arithmetic-place admission, Archimedean completion, and receiver-aperture change become three directions of one system, so their mixed incidence and curvature can be stated without treating any one coordinate plane as absolute.

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I The present question

The problem is not to attach another visualization, statistic, or classifier to prime data. It is to define how an arithmetic ecology changes when:

- several receivers participate in one event but cannot necessarily vary independently;
- their boundaries move, intersect, split, or lose rank;
- a prime place becomes an active axis;
- a formulation changes while its returned invariant remains fixed; and
- the order of these changes may or may not matter.

The existing work already supplied each ingredient separately. Receiver atlases retained co-present perspectives. Prime observations retained exact wheel refinements and valuation lineages. The formulation atlas retained nonidentical paths to π and e . Differential geometry supplied connections, discriminants, and holonomy. What remained absent was a common carrier in which those are directions of one dynamic system.

This paper supplies that carrier. It changes no Soma source and performs no new machine experiment.

I.1 Claim grades

Grade	Use here
Project definition	The receiver-configuration ecology and graded arithmetic accessibility current name the exact data which a situated application must supply.
Standard mathe- matics	Incidence spaces, stratifications, implicit continuation, connections, chain groups, valuation vectors, finite differences, exact series, and prime factorization retain their ordinary meanings.
Derived identity	The stratified transport formula, the separation of mixed incidence from curvature, and the prime-cut calculus follow exactly from those carriers.
Bounded exact wit- ness	The interval 113 to 127 and the twelve-transition arctangent cells are finite exact populations already recorded by laboratory observations.
Open global conse- quence	No theorem below proves RH. It gives the missing configuration space in which the arithmetic, Archimedean, and receiver directions of a proposed proof can be compared without inserting an absolute frame.

II The ecological base

II.1 Actual co-presence precedes coordinates

Definition 1 — Receiver-relative configuration calculus Let the actual co-presence of m receiver lineages be an incidence object

$$E \rightarrow B_1 \times \cdots \times B_m,$$

not the full Cartesian product. Let $\pi_X : X \rightarrow E$ be the family of contemporary configurations, with fiber X_e , and let

$$q_e : X_e \rightarrow \prod_{i=1}^m Y_{i,e} \quad (\text{JOINT RECEIVER})$$

be the joint receiver map. Its image contains only faces which arise together from one actual configuration.

A finite family of typed comparison laws

$$b_a(e, q_{e(x)}) \mathbin{\boxtimes}_a 0, \quad a \in A,$$

defines the receiver-relative accessible region

$$\Omega_e = \left\{ x \in X_e : b_a(e, q_{e(x)}) \mathbin{\boxtimes}_a 0 \text{ for every active } a \right\}. \quad (\text{ACCESSIBLE REGION})$$

Equalities $b_a = 0$ are its oriented boundary sheets. Their sign vectors stratify the total configuration ecology

$$\mathcal{X} = \{(e, x) : e \in E, x \in X_e\}.$$

If $A_{e,x} = \left\{ a : b_a(e, q_{e(x)}) = 0 \right\}$, the receiver discriminant is

$$\Delta = \left\{ (e, x) : \text{rank } D_{x(b_a \circ q_e)}|_{a \in A_{e,x}} < |A_{e,x}| \right\}. \quad (\text{DISCRIMINANT})$$

Thus a smooth chamber is a region where the active law class and the transverse boundary incidence continue. A seam is a crossing of a boundary or of Δ ; it is not inferred from a plotted kink alone.

A “partial change of receiver i ” at e exists only when there is an admissible tangent $V_i \in T_e E$ satisfying

$$D\tau_{j(V_i)} = 0 \quad \text{for every } j \neq i. \quad (\text{RELATIVE PARTIAL})$$

Entanglement may make this subspace empty. If an active square boundary law $B(e, x) = 0$ has an admitted horizontal fiber $H_{e,x}$ for which $(D_x B)|_H$ is invertible, its continuation along V_i is determined by

$$D_x B \nabla_{V_i} x = -(D_e B)[V_i]. \quad (\text{BOUNDARY CONTINUATION})$$

On an affine chart with four distinct marks, the projective Swing

$$\chi(A, B; C, D) = \frac{(A - C)(B - D)}{(A - D)(B - C)}$$

has the exact logarithmic differential

$$\frac{d\chi}{\chi} = \frac{dA - dC}{A - C} + \frac{dB - dD}{B - D} - \frac{dA - dD}{A - D} - \frac{dB - dC}{B - C}. \quad (\text{SWING DIFFERENTIAL})$$

This covector annihilates tangents to the common projective rechart orbit. It therefore measures relative four-member motion, not motion against an absolute coordinate frame.

Boundary. The comparison laws, incidence object, admissible tangents, and transport are supplied by the ecology being studied. The definition does not assert one ambient space, one master clock, or independently movable receivers. A singular discriminant point can require a correspondence, blow-up, or discrete continuation rather than a derivative. The Swing differential is valid only away from coincident marks and the chosen quotient pole; the undivided projective pair remains the complete carrier at a chart boundary.

The first formula is not a state vector. It is the incidence

$$E \rightarrow B_1 \times \cdots \times B_m.$$

The product addresses possible local receiver positions. The object E contains the positions which actually occur together, including plural occurrences with the same addressed tuple. Completing E to the product would assert combinations which the ecology never supplied.

This distinction controls the calculus. A derivative with respect to receiver i is not created by writing a coordinate subscript. It requires an actual tangent in E which varies that receiver while fixing the others. When every admissible tangent changes several receivers together, only that joint direction exists.

The same principle holds in a discrete ecology. There, the tangent is replaced by an admitted neighboring event or correspondence. Adding an unobserved one-coordinate successor would be the discrete equivalent of a false infinitesimal.

II.2 The accessible region is generated, not ambient

At one incidence e , the configuration fiber X_e is received jointly:

$$q_e : X_e \rightarrow \prod_i Y_{i,e}.$$

Typed comparisons generate a region:

$$\Omega_e = \left\{ x : b_{a(e, q_e(x))} \not\propto_a 0 \text{ for all active } a \right\}.$$

The geometry is therefore relative in a precise sense. The receiver, the comparison, and the configuration participate in the definition of Ω_e . There is no need to place them inside an observer-independent rendering box first.

Each equality $b_a = 0$ is an oriented boundary sheet. Intersections of these sheets make corners and higher codimension faces. Where their differentials lose the expected rank, the existing local coordinate law no longer continues regularly. That discriminant is the exact candidate for a FOUNDing or phase seam. A visible kink may testify about such a seam, but it does not define one.

II.3 Basis, unit, and normalization

An axis is a typed local fiber; a unit is a nonzero local frame in it. If $v = [v]_u u$ and the receiver chooses $u' = v$, then

$$[v]_{u'} = 1.$$

This is why a nonzero length can always have coordinate 1 in a receiver normalized to that length. The transformation $u \rightarrow u'$ retains the scale which the reported 1 omits.

Normalization is consequently not an assertion that all lengths are equal. It is a rebase in a local comparison. Orthonormalization is the corresponding construction for a nondegenerate inner-product frame; it owes that inner product and can fail or change species in an indefinite or singular fiber.

II.4 The Swing differentiates relation

For four ordered marks, the cross-ratio Swing is projectively invariant. Its differential is

$$\frac{d\chi}{\chi} = \frac{dA - dC}{A - C} + \frac{dB - dD}{B - D} - \frac{dA - dD}{A - D} - \frac{dB - dC}{B - C}.$$

Every term compares a difference of motions with a difference of positions. Common projective recharting cancels. The expression therefore detects the relative change of the four-member configuration.

This gives the Swing a precise role in a dynamic ecology. It is not a universal update rule imposed on every system. It is the projective differential available whenever four ordered incidences furnish its marks. Another application may supply a different invariant, but it enters the same receiver-relative calculus.

III One calculus for flow and event

III.1 Smooth continuation and discrete wall crossing

Theorem 2 — Receiver-stratified fundamental theorem of transport Let $\gamma : [0, 1] \rightarrow E$ be an oriented receiver path meeting finitely many seams

$$0 < t_1 < \dots < t_r < 1$$

and remaining in smooth transport chambers between them. Let $\mathfrak{S}_t$ be a section of a chain or module bundle along γ , let $\mathcal{P}_{u \rightarrow v}$ be its chamber transport, and let J_k be the declared seam identification from the left fiber to the right fiber. Define the exact jump

$$\Delta_k \mathfrak{S} = \mathfrak{S}_{t_k^+} - J_k \mathfrak{S}_{t_k^-}.$$

Then, in the final fiber,

$$\mathfrak{S}_1 - \mathcal{P}_{0 \rightarrow 1} \mathfrak{S}_0 = \int_0^1 \mathcal{P}_{t \rightarrow 1} \nabla_{\dot{\gamma}(t)} \mathfrak{S}_t dt + \sum_{k=1}^r \mathcal{P}_{t_k^+ \rightarrow 1} \Delta_k \mathfrak{S}. \quad (\text{STRATIFIED FTC})$$

The integral is taken chamber by chamber. When the path is wholly discrete, it is absent and the jump sum is the exact telescoping successor law. When no seam is crossed, the formula reduces to the covariant fundamental theorem.

Demonstration. Apply the transported fundamental theorem separately on every closed chamber segment. Transport each resulting equality into the final fiber. Adjacent one-sided boundary terms cancel after insertion of the declared seam map J_k ; the uncanceled difference is precisely $\Delta_k \mathfrak{S}$. Summing the finite segment identities leaves the two endpoint terms, the transported chamber integrals, and the transported jumps displayed above.

Boundary. The theorem does not manufacture a seam identification when the ecology supplies none. Accumulating infinitely many seams requires a convergence theorem. A scalar count has a distributional derivative supported on wall crossings, but that count forgets which chains crossed and how. Applied to the graded arithmetic accessibility current, the theorem is exact only after its prime aperture, occurrence chain, and receiver boundary are fixed.

The theorem decomposes exact change into two terms:

$$\text{change} = \text{transported chamber flow} + \text{transported seam jumps}.$$

Neither term is privileged. A smooth deformation contributes through a covariant derivative. A quantized boundary event contributes through an exact difference. A mixed ecology carries both.

The formula also clarifies measurement over orders of time. The endpoint difference is not attributed to one instantaneous state. It is factored through the entire oriented path, including each chamber current and each seam. When a lower constituent no longer appears in the transported successor, it may depart; the equality retains its consequence rather than a perfect copy of its former interior.

III.2 Lambda as evaluation of a supplied ecology

Package the local data as

$$\mathcal{K} = (E, \pi_X, q, b, \nabla, J, \nu).$$

Here E is actual co-presence, q the joint receiver, b the boundary family, ∇ chamber transport, J seam conduct, and ν an optional decoration such as prime valuation. For a path γ , define

$$\Lambda_{\mathcal{K}}(\gamma, S_0) = \mathcal{P}_{0 \rightarrow 1} S_0 + \int_0^1 \mathcal{P}_{t \rightarrow 1} \nabla_{\dot{\gamma}} S_t dt + \sum_k \mathcal{P}_{t_k^+ \rightarrow 1} \Delta_k S.$$

By the stratified fundamental theorem this is S_1 .

This is the disciplined sense in which the calculus is universal. Lambda is not a content-free law which guesses the ecology. It evaluates the exact incidence, boundary, transport, and seam data the ecology supplies. The same construction can receive arithmetic, geometry, a physical simulation, an algorithm, language, or a biological network without asserting that those domains share one vocabulary or one domain equation.

IV Second-order ecology

IV.1 Co-presence is not curvature

Theorem 3 — Joint incidence and curvature are the symmetric and ordered second faces

Consider an admitted receiver square with configurations $S_{00}, S_{10}, S_{01}, S_{11}$ carried into one comparison fiber. Its joint second difference is

$$H_{ij}S = S_{11} - S_{10} - S_{01} + S_{00}. \quad (\text{JOINT INCIDENCE})$$

This is the exact inclusion–exclusion coefficient of the feature which appears only when receiver changes i and j are co-present.

If the two paths carry ordered transports

$$U_{00,10}, U_{10,11}, U_{00,01}, U_{01,11},$$

their curvature residual is

$$\mathcal{R}_{ij} = U_{10,11}U_{00,10} - U_{01,11}U_{00,01}. \quad (\text{ORDERED RESIDUAL})$$

With invertible transports this residual may equivalently be represented by returned holonomy. In the smooth limit it becomes

$$R(V_i, V_j) = [\nabla_{V_i}, \nabla_{V_j}] - \nabla_{[V_i, V_j]}. \quad (\text{CURVATURE})$$

The two second faces are logically independent. A commuting square may have $\mathcal{R}_{ij} = 0$ while $H_{ij}S \neq 0$: joint interaction can be present even when its two orders agree. Conversely, nonzero curvature records order-dependent transport and need not create a new scalar population.

If Δ_p is a declared arithmetic place-admission or valuation-layer difference and ∇_{V_i} is an admissible receiver derivative, then

$$\mathcal{C}_{i,p} = \nabla_{V_i}\Delta_p - \Delta_p\nabla_{V_i} \quad (\text{MIXED CURVATURE})$$

measures whether changing the receiver transports the arithmetic cut. It is the precise comparison between “move the perspective, then admit the prime face” and “admit the prime face, then move the perspective.”

Demonstration. Expanding the four values of a finite bivariate difference gives (“JOINT INCIDENCE”) by inclusion–exclusion. Subtracting the two composites around the square gives (“ORDERED RESIDUAL”) by definition. The infinitesimal commutator formula is the standard curvature of a connection after correction by the bracket of the varying vector fields. The examples $S_{ab} = ab$ and two identity transports give $H_{ij}S = 1$ but zero ordered residual, proving that neither second face can replace the other. The mixed formula is the same commutator with one continuous receiver direction and one discrete arithmetic direction.

Boundary. A receiver “partial” is undefined when the incidence base does not admit its independent tangent. In that case one must use an actual path or correspondence, not insert a false coordinate direction. The mixed curvature is relative to a specified arithmetic cut and comparison transport; it is not a universal statistic of a prime.

For a two-direction square, the mixed finite difference

$$H_{ij}S = S_{11} - S_{10} - S_{01} + S_{00}$$

is the part not attributable to either direction alone. It is an exact incidence term. If $S_{ab} = ab$, then $H_{ij}S = 1$: the joint feature exists only when both inputs are present.

Curvature asks another question. Transport first along i and then j ; transport first along j and then i ; compare the two returns. Their difference is the ordered residual

$$\mathcal{R}_{ij} = U_{10,11}U_{00,10} - U_{01,11}U_{00,01}.$$

The two paths may agree while the mixed incidence remains nonzero. This is why $\partial^2 = 0$, a commuting diagram, or a vanishing curvature does not mean that second-order interaction disappeared.

IV.2 Receiver–prime mixed curvature

Let Δ_p admit or resolve a prime-place boundary and let V_i be an admissible receiver direction. Then

$$\mathcal{C}_{i,p} = \nabla_{V_i}\Delta_p - \Delta_p\nabla_{V_i}$$

asks whether those operations commute.

- If $\mathcal{C}_{i,p} = 0$, the prime cut transports naturally through that receiver motion.
- If it is nonzero, the receiver change alters which arithmetic face the cut exposes, or the arithmetic cut alters the receiver's continuation.
- Even when it vanishes, the symmetric mixed difference can record co-present receiver–prime incidence.

This is the higher-order vector calculus requested by the laboratory. Its directions are not axes of an absolute ambient space. They are admitted motions and discrete differences in the actual incidence ecology.

V The arithmetic field

V.1 Valuation fibers retain composition

Definition 4 — Graded arithmetic accessibility current Let C_e be an oriented locally finite chain of arithmetic occurrences in the accessible configuration Ω_e . Let P_e be its declared finite prime aperture and let

$$\nu_e : C_e \rightarrow \mathbb{N}^{P_e}$$

assign the exact prime-valuation address of every arithmetic occurrence on which that address is defined. For $\alpha \in \mathbb{N}^{P_e}$, write

$$C_{\alpha,e} = C_e|_{\Omega_e \cap \nu_e^{-1}(\alpha)}.$$

The **graded arithmetic accessibility current** is

$$\mathfrak{A}_{e(X)} = \sum_{\alpha \in \mathbb{N}^{P_e}} [C_{\alpha,e}] X^\alpha, \quad X^\alpha = \prod_{p \in P_e} X_p^{\alpha_p}. \quad (\text{ACCESSIBILITY CURRENT})$$

Its coefficients are chains or signed occurrence populations, not scalar probabilities. Orientation, multiplicity, boundary, and source incidence remain available before any augmentation.

Total valuation degree gives

$$\mathfrak{A}_e = \sum_{d \geq 0} \mathfrak{A}_e^{(d)}, \quad \mathfrak{A}_e^{(d)} = \sum_{|\alpha|_1 = d} [C_{\alpha,e}] X^\alpha.$$

Degree one contains the primitive integer-prime axes X_p . Degree two separates the two nonidentical faces

$$X_p^2 \quad \text{and} \quad X_p X_q \quad (p \neq q),$$

so prime squares and distinct-prime semiprimes are not collapsed.

A declared augmentation

$$\varepsilon : C_{\bullet}(\Omega_e; \mathbb{Z}) \rightarrow \mathbb{Z}$$

may later return signed cardinality, mass, or another exact census:

$$Z_{e(\mathbf{X})} = \varepsilon(\mathfrak{A}_{e(\mathbf{X})}).$$

That quotient is a receiver of the current. It is not the current's identity.

Boundary. Integer valuation is one lawful decoration of a general configuration, not a universal taxonomy. The prime aperture may grow, and an occurrence outside it retains an unresolved valuation residual rather than being assigned a false zero coordinate. If the arithmetic objects live in a ring other than the positive integers, its units and prime species must be declared anew. Infinite sums require a topology or formal-series interpretation; every bounded cell in this definition is exact.

The current

$$\mathfrak{A}_{e(\mathbf{X})} = \sum_{\alpha} [C_{\alpha, e}] \mathbf{X}^{\alpha}$$

retains what a scalar “number of accessible primes” cannot. Each coefficient is the oriented population occupying one valuation address. Its degree-one part contains primitive prime axes. Its degree-two part distinguishes

$$X_p^2 \quad \text{from} \quad X_p X_q.$$

The first is repeated traversal of one prime axis; the second is a crossing of two distinct prime axes.

The formal variables do not claim that primes are spatial coordinate axes. They address exact multiplicative directions supplied by unique factorization. A geometric receiver may later embed those addresses into a simplicial complex, logarithmic chart, circle, conic, or another surface. That embedding is a face of the valuation current and owes its own transport law.

V.2 Probability is a later receiver

Applying an augmentation ε gives

$$Z_{e(\mathbf{X})} = \varepsilon(\mathfrak{A}_{e(\mathbf{X})}).$$

One may then normalize a declared finite measure and form conditional recurrence ratios. That probability face is exact in its event algebra, but it is downstream of the occurrence current. It does not turn the deterministic wall crossings into ontological chance.

Likewise, loss is an oriented residual between two transported currents. Its sign refers to the chosen comparison orientation. It is not reward, punishment, or a scalar property attached to one arithmetic object.

V.3 Prime admission is a boundary operation

Theorem 5 — Prime admission has an exact cut, Hessian, and valuation-layer decomposition Let

$$\mathcal{C}_S = \left\{ n \in \mathcal{N} : \gcd\left(n, \prod_{q \in S} q\right) = 1 \right\}$$

be a finite situated candidate chain inside a declared aperture $\mathcal{N}$. For $p \notin S$, let K_p restrict this chain to $\mathcal{C}_{S \cup \{p\}}$. In the common enlarged valuation ring, define

$$\Delta_p \mathfrak{A}_S = K_p \mathfrak{A}_S - \mathfrak{A}_S.$$

Then

$$\Delta_p \mathfrak{A}_S = - \sum_{n \in \mathcal{C}_S, p|n} [n] \mathbf{X}^{\nu(n)}. \quad (\text{FIRST CUT})$$

No removed occurrence is anonymous: its complete valuation monomial remains the coefficient address of the wall crossing.

For distinct $p, q \notin S$, static sieve restrictions commute and

$$\Delta_q \Delta_p \mathfrak{A}_S = \Delta_p \Delta_q \mathfrak{A}_S = \sum_{n \in \mathcal{C}_S, pq|n} [n] \mathbf{X}^{\nu(n)}. \quad (\text{CUT HESSIAN})$$

Hence the cut commutator vanishes while the joint second difference may be nonzero. The degree-two projection of (“CUT HESSIAN”) isolates the mixed semiprime face $X_p X_q$.

Prime squares are not obtained by pretending to admit p twice. They belong to the pairwise-disjoint valuation layers

$$\mathcal{C}_{S,k} = \{n \in \mathcal{C}_S : v_{p(n)} = k\}, \quad \mathcal{C}_S = \cup_{k \geq 0} \mathcal{C}_{S,k},$$

whose degree-two $k = 2$ coefficient is X_p^2 . Thus the exact second-order arithmetic geometry consists of both cross-axis incidence $X_p X_q$ and repeated traversal X_p^2 .

Demonstration. The first identity is the chain difference between a population and its restriction away from multiples of p . Applying the q difference restores with positive sign precisely the terms removed by both cuts. Divisibility by distinct primes is order-independent, proving the equality of the two mixed differences and the vanishing commutator. Unique factorization grades the shared population by valuation degree; degree two has exactly the partitions $2e_p$ and $e_p + e_q$.

Boundary. Commutation holds for static exact divisibility cuts in one fixed aperture. A receiver-dependent aperture, transported coefficient law, or changing arithmetic domain may have nonzero mixed curvature as defined above. The first and second differences are chain identities; replacing them by cardinalities discards location, order, and valuation species.

A new place p cuts out the occurrences divisible by p . The first difference is negative because those occurrences cross out of the survivor region. Admitting distinct p and q in either order gives the same static sieve population, so the cut commutator vanishes. Their mixed second difference is nevertheless the complete population divisible by pq .

This is a concrete arithmetic instance of the second-order distinction:

$$\text{zero cut curvature} \neq \text{zero joint incidence}.$$

Prime squares require the valuation layer $v_p = 2$; applying the same idempotent admission twice would erase that distinction.

VI The interval 113 to 127

Theorem 6 — The interval from 113 to 127 has a six-occurrence arithmetic wall current In the closed integer aperture $[113, 127]$, begin with the 2-coprime population and admit the successive sieve axes 3, 5, 7, 11. The four exact first-cut currents are

$$\Delta_3 \mathfrak{A} = -(X_3^2 X_{13} + X_3 X_{41}),$$

$$\Delta_5 \mathfrak{A} = -(X_5 X_{23} + X_5^3),$$

$$\Delta_7 \mathfrak{A} = -X_7 X_{17}, \quad \Delta_{11} \mathfrak{A} = -X_{11}^2. \quad (\text{INTERVAL WALL CURRENT})$$

They correspond respectively to

$$117 = 3^2 \cdot 13, \quad 123 = 3 \cdot 41, \quad 115 = 5 \cdot 23, \quad 125 = 5^3, \quad 119 = 7 \cdot 17, \quad 121 = 11^2.$$

The endpoint survivors are 113 and 127.

Consequently, the final scalar gap 14 is the augmentation of a wall current containing three mixed degree-two crossings, one prime-square degree-two crossing, and two degree-three crossings. The successive surviving place sets

$$\{2\}, \{2, 3\}, \{2, 3, 5\}, \{2, 3, 5, 7\}, \{2, 3, 5, 7, 11\}$$

determine the exact refinement lineage.

Demonstration. The odd population is

$$113, 115, 117, 119, 121, 123, 125, 127.$$

Cutting by 3 removes 117, 123; cutting the survivors by 5 removes 115, 125; the 7 cut removes 119; and the 11 cut removes 121. The displayed factorizations are exact and unique. Projection by total valuation degree gives the stated four degree-two and two degree-three crossings. Only 113 and 127 remain, so augmenting their ordered endpoint difference gives 14 while forgetting the six interior wall events.

Boundary. This theorem grades one finite situated interval. It neither predicts the next prime nor identifies one universal metric between primes. Its purpose is to expose the topology hidden by the endpoint quotient: arithmetic accessibility changes through typed valuation walls.

The familiar refinement table is now a single path in configuration space:

New axis	Crossing occurrences	Valuation current
3	117, 123	$-(X_3^2 X_{13} + X_3 X_{41})$
5	115, 125	$-(X_5 X_{23} + X_5^3)$
7	119	$-X_7 X_{17}$
11	121	$-X_{11}^2$

The final gap $127 - 113 = 14$ is correct, but it is not the identity of this path. It is an endpoint quotient after six interior arithmetic boundary events. The exact topology includes:

- three distinct-prime semiprime crossings;
- one prime-square crossing; and
- two degree-three crossings.

This is what it means to ask how the accessible prime and semiprime population changes with a receiver region. The answer is a graded wall current. A count can be read from it, but the count alone cannot reconstruct the current.

VII Transcendental invariants as formulation ecologies

VII.1 Evaluation fibers replace “the formula”

Theorem 7 — A transcendental evaluation fiber carries nonidentical arithmetic accessibility currents For a declared exact evaluation law

$$\text{ev} : \mathcal{F} \rightarrow \mathcal{V},$$

the fiber

$$\mathcal{E}_\tau = \text{ev}^{-1}(\tau)$$

contains the formulations returning the invariant τ . Its members need not have one arithmetic accessibility current. A proved formulation transport $T : F \rightarrow G$ has the exact residual

$$\text{Res}_T = T_* \mathfrak{A}_F - \mathfrak{A}_G. \quad (\text{FORMULATION RESIDUAL})$$

Vanishing means that this declared transport preserves the decorated arithmetic current. Nonvanishing records entering, departing, or rebased axes without denying that F and G return the same τ .

The exponential specialization of a valuation address is

$$\Phi_{s(\alpha)} = \prod_p p^{-s\alpha_p} = \exp\left(-s \sum_p \alpha_p \log p\right). \quad (\text{EXPONENTIAL SPECIALIZATION})$$

For the factorial series of e ,

$$v_{p((n+1)!)} - v_{p(n!)} = v_{p(n+1)},$$

so each successor transports the complete valuation current of $n + 1$ into the next denominator.

For a real rational $x = \frac{u}{v}$ in lowest terms with $v \neq 0$ and $|u| \leq |v|$, let

$$A(x) = \sum_{n \geq 0} (-1)^n \frac{x^{2n+1}}{2n+1}.$$

This is the exact arctangent series, conditionally at its endpoints. Its exact term transport is

$$\frac{a_{n+1}}{a_n} = -\frac{u^2(2n+1)}{v^2(2n+3)},$$

hence

$$\text{div}(a_{n+1}) - \text{div}(a_n) = 2\text{div}(u) - 2\text{div}(v) + \text{div}(2n+1) - \text{div}(2n+3). \quad (\text{ARCTANGENT CURRENT})$$

If an odd prime p divides v , collisions with the moving odd current lie on the exact modular sheets

$$2n+1 \equiv 0 \pmod{p^k} \quad \text{or} \quad 2n+3 \equiv 0 \pmod{p^k}. \quad (\text{COLLISION SHEETS})$$

Thus the Machin presentation

$$\pi = 16A\left(\frac{1}{5}\right) - 4A\left(\frac{1}{239}\right)$$

has parameter axes 5, 239, while its Gaussian norm closure uses

$$1 + 5^2 = 2 \cdot 13, \quad 1 + 239^2 = 2 \cdot 13^4.$$

These are two related receiver faces, not one fixed “prime support of π .” In the first twelve term transitions, the $\frac{1}{5}$ arm crosses two 5-collision sheets and cancels 5^2 internally; the $\frac{1}{239}$ arm crosses no 239-sheet. The fixed Chudnovsky scale similarly has exact support

$$640320 = 2^6 \cdot 3 \cdot 5 \cdot 23 \cdot 29.$$

Demonstration. The residual is the definition of comparing two decorated currents after transport. The exponential identity follows by substituting the valuation factorization into n^{-s} . The factorial identity follows from $(n+1) \neq (n+1)n!$. Dividing consecutive arctangent terms gives (“ARCTANGENT CURRENT”); taking prime valuations gives the collision congruences. The two Gaussian norm factorizations and the factorization of 640320 are direct integer identities. Solving the displayed congruences in the bounded twelve-transition aperture gives the two 5-sheet crossings and no 239-sheet crossing recorded by the exact presentation-topology cell.

Boundary. Equal evaluation is not a license to identify formulations, and unequal coefficient support is not a contradiction. The bounded collision statement is exact only in its named aperture. This theorem does not assign one universal prime taxonomy to a transcendental invariant or infer the Riemann zero set from coefficient supports. It supplies the exact mixed arithmetic/formulation direction which such a global transport theorem would have to use.

For an invariant τ , the fiber

$$\mathcal{E}_\tau = \text{ev}^{-1}(\tau)$$

is the population of exact formulations returning it. This does not identify their algorithms, coefficients, convergence laws, or internal arithmetic currents. A proved transformation $T : F \rightarrow G$ carries those decorations; its residual

$$\text{Res}_T = T_* \mathfrak{A}_F - \mathfrak{A}_G$$

states exactly what enters, departs, or changes basis.

This makes a transcendental number an invariant face of a dynamic formulation ecology. The invariant is not reduced to one series, and the series is not dismissed because other series return the same invariant.

VII.2 The e-face

The exponential carries additive valuation length into multiplicative amplitude:

$$\alpha \mapsto \prod_p p^{-s\alpha_p} = \exp \left(-s \sum_p \alpha_p \log p \right).$$

For the factorial series of e , the exact successor law

$$v_{p((n+1)!)} - v_{p(n!)} = v_{p(n+1)}$$

says that the arithmetic composition of $n + 1$ becomes the next denominator current. The familiar differential identity $d \frac{\exp(kx)}{dx} x = k \exp(kx)$ is the corresponding chain-rule face: the outer invariant form is retained while the parameter trajectory contributes its own rate.

VII.3 The pi-face

For the arctangent series, a reduced real rational parameter $x = \frac{u}{v}$ with $v \neq 0$ and $|u| \leq |v|$ supplies a persistent scale current while the odd denominator advances:

$$\frac{a_{n+1}}{a_n} = -\frac{u^2(2n+1)}{v^2(2n+3)}.$$

The moving odd face and fixed parameter face meet on modular sheets. This explains why $A(\frac{1}{5})$ and $A(\frac{1}{239})$ can instantiate the same series law but have different local cancellation topologies.

Machin's formula exposes several exact faces at once:

Parameter face	5, 239
Gaussian norm face	$1 + 5^2 = 2 \cdot 13$ and $1 + 239^2 = 2 \cdot 13^4$
Moving series face	$2n + 1$ and $2n + 3$ with their prime valuations
Returned invariant	π after the exact arctangent identity

None is “the prime support of π .” They are related projections of one formulation path. Chudnovsky supplies another ecology whose fixed scale

$$640320 = 2^6 \cdot 3 \cdot 5 \cdot 23 \cdot 29$$

has a different valuation face while returning the same invariant.

VIII Relation to zeta and RH

VIII.1 Three directions of one incidence system

The earlier RH work repeatedly encountered three frames:

Direction	Elementary action	Exact face
Arithmetic place	admit or resolve p	valuation strata, Möbius cut, Euler factor, prime-power current
Archimedean/formulation	Fourier, Poisson, Mellin, Gamma, series rebase	scale, boundary normalization, completed analytic current
Receiver aperture	move, refine, intersect, or reorient a boundary	accessible chain, crossing order, seam jump, returned holonomy

These are not three unrelated metaphors. They are three admissible directions in a single configuration family. The finite explicit formula already demonstrates that prime-power current and completed zero current are two receivers of one exact contour-transport identity. The present calculus adds the missing question:

How does that identity change under prime admission and receiver motion?

Its answer consists of mixed Hessians and mixed curvature, not a scalar correlation.

VIII.2 Why one half appears

The completed zeta involution exchanges

$$s \rightarrow 1 - \bar{s}.$$

Writing $s = \frac{1}{2} + z$ turns it into $z \rightarrow -\bar{z}$. The line $\text{Re } s = \frac{1}{2}$ is therefore the fixed seam of an exact two-pole exchange. The value $\frac{1}{2}$ is not declared positive in isolation. It is the midpoint coordinate determined by that involution after the two exchanged poles 0 and 1 are used as the frame.

Changing coordinates can reverse a reported sign, but it cannot remove the fixed-set relation if the involution is transported naturally. This is the appropriate receiver-relative meaning of the critical line.

VIII.3 What “positivity” can mean here

In a declared Hermitian receiver, a returned form Q is nonnegative when

$$Q(f, f) \geq 0$$

for every admitted test current f , using that receiver’s ordered real scalar field and orientation. This is not an absolute adjective attached to a zero or a prime. Under an invertible rechart g , the congruent response

$$Q'(gf, gf) = Q(f, f)$$

retains the sign. Reversing the response law itself creates a different oriented comparison.

One classical RH route asks for exactly such a nonnegative completed prime/zero response for every admissible test function. The present calculus does not assume it. It identifies the structure from which it would have to be derived: the jointly transported arithmetic, Archimedean, and aperture current, including all seam terms. A proposed factorization must commute with those three directions; otherwise its sign is only a chart-local artifact.

VIII.4 The proof-bearing use of the calculus

Let Δ_p denote place admission, ∇_A Archimedean/formulation transport, and ∇_R receiver-aperture transport. A complete RH construction in this language must determine, on an exact admissible test-current space:

$$[\nabla_A, \Delta_p], \quad [\nabla_R, \Delta_p], \quad [\nabla_R, \nabla_A],$$

together with their symmetric mixed-incidence terms and seam jumps.

There are then two lawful outcomes for each cell:

1. it commutes, supplying a natural transport of the prime-place current through completion and receiver change; or
2. it does not commute, and its exact residual becomes part of the completed response rather than being omitted.

This is stronger than searching finite examples and more precise than asking for a favorable perspective. It asks for a global naturality or an explicit residual on every admitted path. It also explains why the π and e formulation ecologies matter: Gamma, Gaussian, Fourier, Mellin, logarithmic, and exponential transports are the Archimedean paths through which the finite-prime current reaches the completed receiver.

IX Universality without an absolute world

The calculus has one reusable pattern:

$$\text{actual incidence} \rightarrow \text{receiver-generated region} \rightarrow \text{transport plus seams} \rightarrow \text{changed current}.$$

Its content is supplied locally.

- In arithmetic, valuation axes decorate integer compositions.
- In differential geometry, tangent directions and connections transport local fibers.
- In an algorithm, states and transitions form the causal path while branches and halting conditions form boundaries.
- In a physical model, units, constitutive laws, conserved currents, and boundary conditions instantiate the same carrier.
- In an information ecology, inscriptions, transducers, and consequences determine which distinctions remain active.

This is not the assertion that every domain is literally one substance or one equation. It is the claim that lawful relation, restriction, transport, boundary, seam, return, and receiver-exact factorization can be expressed in one dependency-preserving mathematical grammar.

X Established result

This cut establishes the following reusable object:

$$\mathcal{K}_{\text{holonic}} = (\text{receiver-relative configuration, graded occurrence current, stratified transport, mixed incidence/curvature}).$$

It resolves the immediate conceptual gap:

- accessible prime and semiprime populations are chain-valued boundary currents, not floating densities;
- receiver partials are actual incidence directions, not assumed axes;
- smooth flow and discrete founding coexist in one exact theorem;

- interaction and path dependence are distinct second-order phenomena;
- π , e , and other invariants are formulation fibers with transported arithmetic ecologies; and
- the RH problem can now be posed on one three-direction configuration system rather than alternating among disconnected arithmetic, geometric, and analytic pictures.

The construction ends here because the keystone object is now defined, catalogued, dependency-ordered, and instantiated by existing exact evidence.